

Настройка DHCP для IPv4 и IPv6 в GNS3

Лабораторная работа №7

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Цель лабораторной работы

Получить практические навыки настройки **DHCPv4**, **DHCPv6 Stateless** и **DHCPv6 Stateful** на маршрутизаторе **VyOS** в среде GNS3.

1. Настройка DHCPv4 и автоматическое получение IPv4-адресов клиентом.
2. Настройка IPv6 и механизмов SLAAC.
3. Организация DHCPv6 в режимах Stateless и Stateful.
4. Анализ трафика DHCP/ICMPv6 в Wireshark.

Теоретическая часть

- Адрес формируется через SLAAC
- DHCP передает только:
 - DNS
 - доменную зону
 - время жизни записей

- Аналог DHCPv4
- Сервер выделяет IPv6-адреса
- Сообщения: Solicit → Advertise → Request → Reply

Топология сети

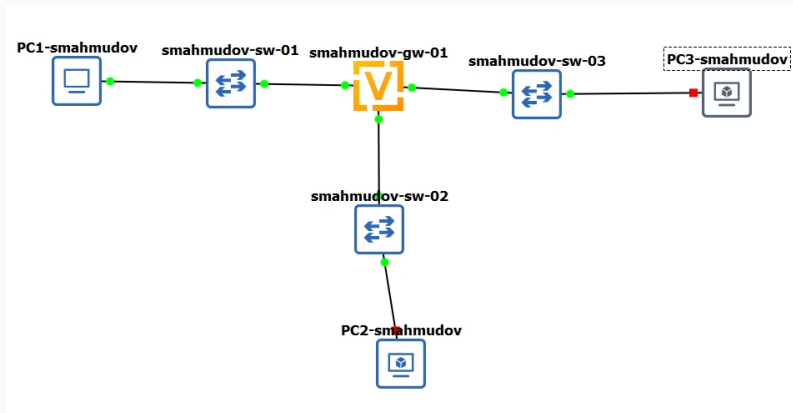
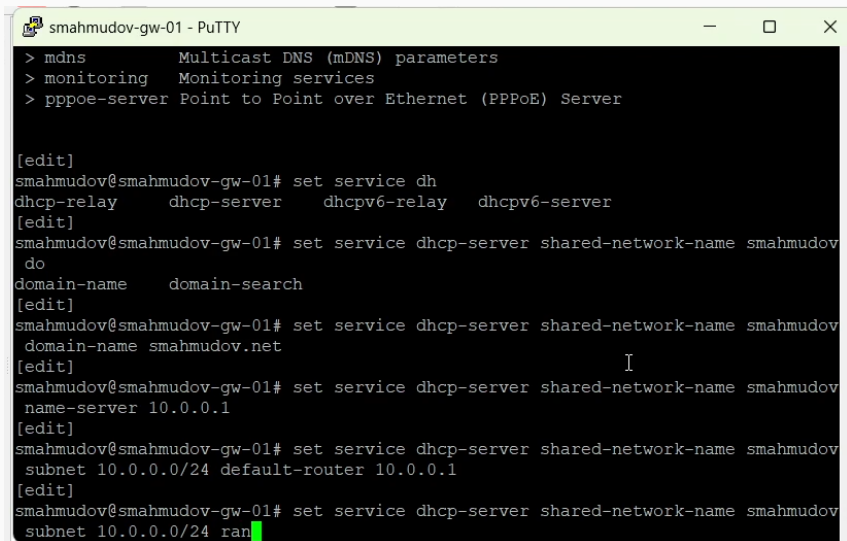


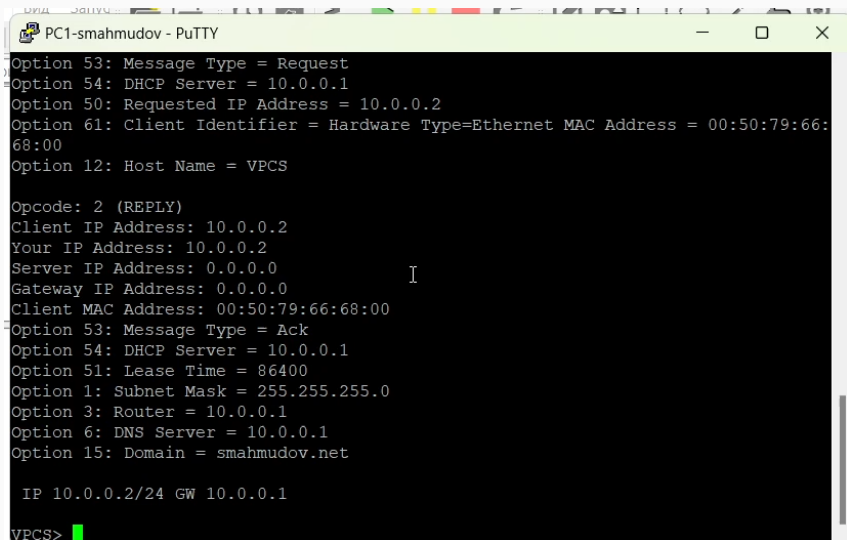
Рис. 1: Топология IPv6-сети



```
smahmudov-gw-01 - PuTTY
> mdns      Multicast DNS (mDNS) parameters
> monitoring Monitoring services
> pppoe-server Point to Point over Ethernet (PPPoE) Server

[edit]
smahmudov@smahmudov-gw-01# set service dh
dhcp-relay      dhcp-server      dhcpv6-relay      dhcpv6-server
[edit]
smahmudov@smahmudov-gw-01# set service dhcp-server shared-network-name smahmudov
do
domain-name      domain-search
[edit]
smahmudov@smahmudov-gw-01# set service dhcp-server shared-network-name smahmudov
domain-name smahmudov.net
[edit]
smahmudov@smahmudov-gw-01# set service dhcp-server shared-network-name smahmudov
name-server 10.0.0.1
[edit]
smahmudov@smahmudov-gw-01# set service dhcp-server shared-network-name smahmudov
subnet 10.0.0.0/24 default-router 10.0.0.1
[edit]
smahmudov@smahmudov-gw-01# set service dhcp-server shared-network-name smahmudov
subnet 10.0.0.0/24 ran
```

Клиент получает IPv4-адрес

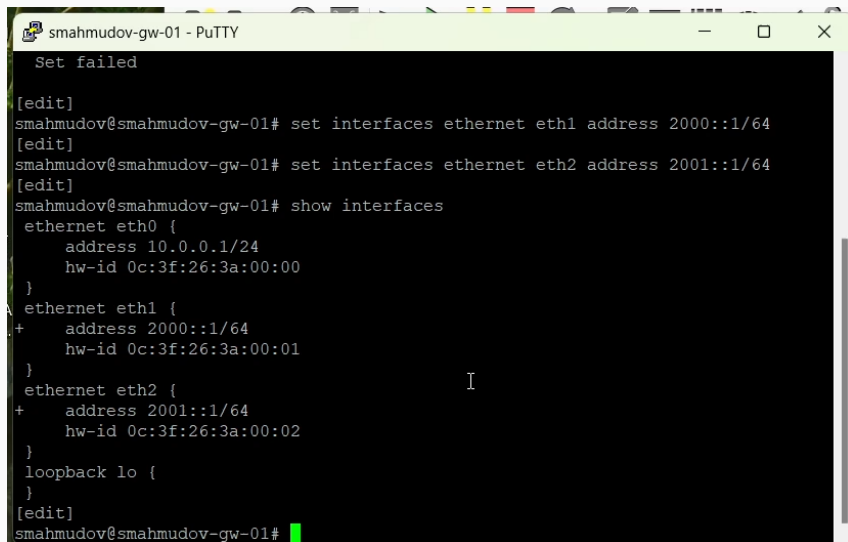


```
PC1-smahmudov - PuTTY
Option 53: Message Type = Request
Option 54: DHCP Server = 10.0.0.1
Option 50: Requested IP Address = 10.0.0.2
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:68:00
Option 12: Host Name = VPCS

Opcode: 2 (REPLY)
Client IP Address: 10.0.0.2
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Ack
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = smahmudov.net

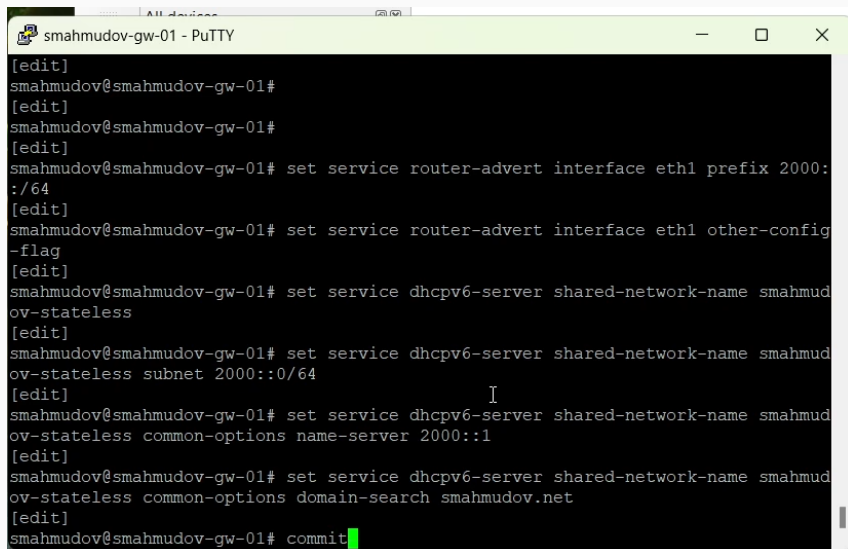
IP 10.0.0.2/24 GW 10.0.0.1
VPCS>
```

Рис. 3: Получение DHCP-адреса



```
smahmudov-gw-01 - PuTTY
Set failed

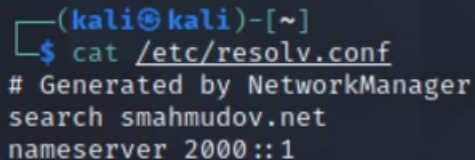
[edit]
smahmudov@smahmudov-gw-01# set interfaces ethernet eth1 address 2000::1/64
[edit]
smahmudov@smahmudov-gw-01# set interfaces ethernet eth2 address 2001::1/64
[edit]
smahmudov@smahmudov-gw-01# show interfaces
  ethernet eth0 {
    address 10.0.0.1/24
    hw-id 0c:3f:26:3a:00:00
  }
  ethernet eth1 {
+   address 2000::1/64
    hw-id 0c:3f:26:3a:00:01
  }
  ethernet eth2 {
+   address 2001::1/64
    hw-id 0c:3f:26:3a:00:02
  }
  loopback lo {
  }
[edit]
smahmudov@smahmudov-gw-01#
```



```
[edit]
smahmudov@smahmudov-gw-01#
[edit]
smahmudov@smahmudov-gw-01#
[edit]
smahmudov@smahmudov-gw-01# set service router-advert interface eth1 prefix 2000:
:/64
[edit]
smahmudov@smahmudov-gw-01# set service router-advert interface eth1 other-config
-flag
[edit]
smahmudov@smahmudov-gw-01# set service dhcpv6-server shared-network-name smahmud
ov-stateless
[edit]
smahmudov@smahmudov-gw-01# set service dhcpv6-server shared-network-name smahmud
ov-stateless subnet 2000::0/64
[edit]
smahmudov@smahmudov-gw-01# set service dhcpv6-server shared-network-name smahmud
ov-stateless common-options name-server 2000::1
[edit]
smahmudov@smahmudov-gw-01# set service dhcpv6-server shared-network-name smahmud
ov-stateless common-options domain-search smahmudov.net
[edit]
smahmudov@smahmudov-gw-01# commit
```

```
(kali㉿kali)-[~]  
$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet6 2000::e2be:ac68:9681:1027 prefixlen 64 scopeid 0x0<global>  
    inet6 fe80::8a5f:79e2:eb4f:2b0a prefixlen 64 scopeid 0x20<link>  
    ether 0c:a3:1f:ea:00:00 txqueuelen 1000 (Ethernet)  
    RX packets 17 bytes 781 (781.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 21 bytes 3152 (3.0 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

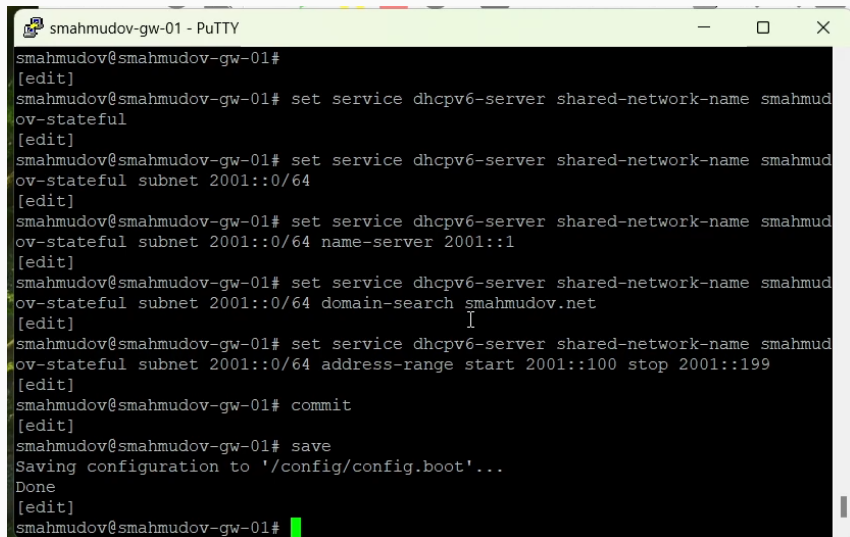
Рис. 6: IPv6 адрес SLAAC



```
(kali㉿kali)-[~]  
$ cat /etc/resolv.conf  
# Generated by NetworkManager  
search smahmudov.net  
nameserver 2000::1
```

Рис. 7: resolv.conf

DHCPv6 Stateful



```
smahmudov-gw-01 - PuTTY
smahmudov@smahmudov-gw-01#
[edit]
smahmudov@smahmudov-gw-01# set service dhcpv6-server shared-network-name smahmud
ov-stateful
[edit]
smahmudov@smahmudov-gw-01# set service dhcpv6-server shared-network-name smahmud
ov-stateful subnet 2001::0/64
[edit]
smahmudov@smahmudov-gw-01# set service dhcpv6-server shared-network-name smahmud
ov-stateful subnet 2001::0/64 name-server 2001::1
[edit]
smahmudov@smahmudov-gw-01# set service dhcpv6-server shared-network-name smahmud
ov-stateful subnet 2001::0/64 domain-search smahmudov.net
[edit]
smahmudov@smahmudov-gw-01# set service dhcpv6-server shared-network-name smahmud
ov-stateful subnet 2001::0/64 address-range start 2001::100 stop 2001::199
[edit]
smahmudov@smahmudov-gw-01# commit
[edit]
smahmudov@smahmudov-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
smahmudov@smahmudov-gw-01#
```

Рис. 8: DHCPv6 Stateful

Клиент PC3 получает адрес DHCPv6

```
RCV: Advertisement recorded.
PRC: Selecting best advertised lease.
PRC: Considering best lease.
PRC: X-- Initial candidate 00:01:00:01:30:af:1d:06:0c:3f:26:3a:00:01 (s: 10105, p: 0).
XMT: Forming Request, 0 ms elapsed.
XMT: X-- IA_NA 54:ab:00:00
XMT: | X-- Requested renew +3600
XMT: | X-- Requested rebind +5400
XMT: | | X-- IAADDR 2001::199
XMT: | | | X-- Preferred lifetime +7200
XMT: | | | X-- Max lifetime +7500
XMT: V IA_NA appended.
XMT: Request on eth0, interval 960ms.
RCV: Reply message on eth0 from fe80::e3f:26ff:fe3a:2.
RCV: X-- IA_NA 54:ab:00:00
RCV: | X-- starts 1763468204
RCV: | X-- t1 - renew +0
RCV: | X-- t2 - rebind +0
RCV: | X-- [Options]
RCV: | | X-- IAADDR 2001::199
RCV: | | | X-- Preferred lifetime 7200.
RCV: | | | X-- Max lifetime 7500.
RCV: X-- Server ID: 00:01:00:01:30:af:1d:06:0c:3f:26:3a:00:01
PRC: Bound to lease 00:01:00:01:30:af:1d:06:0c:3f:26:3a:00:01.
```

Рис. 9: dhclient

```
└─$ ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 2001::199 prefixlen 128 scopeid 0<global>
    inet6 fe80::4456:4ca1:35eb:858c prefixlen 64 scopeid 0<link>
    ether 0c:ff:54:ab:00:00 txqueuelen 1000 (Ethernet)
    RX packets 4 bytes 534 (534.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 54 bytes 8276 (8.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)-[~]
└─$ route -n -A inet6
Kernel IPv6 routing table
```

Destination	Next Hop	Flag	Met	Ref	Use	If
::1/128	::	U	256	2	0	lo
2001::199/128	::	U	256	1	0	eth0
fe80::/64	::	U	100	1	0	eth0
::/0	::	!n	-1	1	0	lo
::1/128	::	Un	0	4	0	lo
2001::199/128	::	Un	0	2	0	eth0
fe80::4456:4ca1:35eb:858c/128	::	Un	0	3	0	eth0
ff00::/8	::	U	256	3	0	eth0
::/0	::	!n	-1	1	0	lo

```
(kali@kali)-[~]
```

Рис. 10: IPv6 настройки

Вывод

- Настроен DHCPv4 на маршрутизаторе VyOS.
- Реализованы SLAAC, Stateless и Stateful DHCPv6.
- Клиенты корректно получают IPv4/IPv6-адреса и сетевые параметры.
- Wireshark подтвердил правильность обмена сообщениями DHCPv4/DHCPv6.
- Сеть полностью функционирует в IPv4 и IPv6-окружении.